Hotel Booking Demand Analysis

Driving Revenue Through Cancellation Insights | Sector: Hospitality & Tourism
Team 6

GitHub: Github

Team:-

- Aditya Verma
- Raghav
- Farhana
- Abhijeet Kumar
- Mukul
- Malhar

Context & Problem Statement

OVERVIEW OF PROJECT SCOPE AND BUSINESS PURPOSE

Booking cancellations are significantly impacting hotel revenue, leading to financial losses and demand uncertainty.

The objective is to identify key drivers of cancellations and recommend data-driven strategies to reduce revenue leakage and improve booking stability.

AI-DRIVEN ANALYTICAL SUMMARY (BASED ON DASHBOARD INSIGHTS)

- Core Business Issue
- Key Drivers of Cancellations
- Deposit Policy Impact
- Lead Time Effect
- Hotel Type Variation
- Pricing Sensitivity (ADR Impact)
- Customer Loyalty Gap
- Business Interpretation
- Decision Direction
- Conclusion

Data Engineering — Source to Pipeline

▶ Source Dataset

- • Name: Hotel Bookings Demand (Kaggle)
- • Size: 119,390 rows × 32 columns
- • Period: July 2015 – August 2017
- • Hotels: City Hotel, Resort Hotel

Kaggle - https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand/data

▶ Python ETL: Major Cleaning Steps:

- **Handling Missing Values in 'children' column**: Filled missing values with 0, assuming no children were present.
- **Handling Missing Values in 'country' column**: Filled missing values with 'Unknown' to retain records without introducing bias.
- **Dropping Columns with Excessive Missing Values**: Dropped agent and company columns due to a large number of missing values (16,340 and 112,593 respectively), making them unreliable for analysis.
- **Data Type Conversion**: Converted the reservation_status_date column to datetime format for proper time-based analysis.
- **Feature Engineering**: Created two new features:
- **total_stay**: Calculated as stays_in_weekend_nights + stays_in_week_nights.
- **revenue**: Calculated as adr * total_stay.

KPI & Metrics Framework

KEY KPIS & THEIR BUSINESS USE

Revenue KPIs

- Total Revenue → Overall business performance
- Revenue Realized → Actual earnings after cancellations
- Revenue Loss → Financial impact of cancellations
- ADR (Avg. Daily Rate) → Pricing effectiveness

Demand & Booking KPIs

- Total Bookings → Demand volume
- Total Guests → Customer scale
- Booking Change Rate (BCR) → Booking instability indicator

Cancellation-Focused KPIs (Core Problem)

- Cancellation Rate → % of lost bookings
- Deposit Type Impact → Customer commitment & cancellation risk
- Lead Time → Early signal of cancellation probability

Customer Behavior KPIs

- Total Repeated Guests → Loyalty & retention strength
- Customer Type → Segment-level behavior differences
- Total Special Requests → Booking intent strength

Operational KPIs

- ALOS (Avg. Length of Stay) → Revenue stability per booking
- Room Assignment Discrepancy → Service quality issues

Computation Note

All KPIs are computed using Python (Pandas) and validated via Tableau dashboards for segment-level

01

CANCELLATIONS ARE A MAJOR REVENUE DRAIN (~37%)

A significant portion of bookings are cancelled, resulting in ~$13M revenue loss, making cancellations the primary business risk.

02

CITY HOTELS FACE SIGNIFICANTLY HIGHER CANCELLATION RISK THAN RESORT HOTELS

Urban properties show ~1.5× higher cancellation rates, indicating demand volatility and lower booking commitment in city segments.

03

LOW-COMMITMENT BOOKINGS (NO DEPOSIT) DRIVE THE MAJORITY OF CANCELLATIONS

Bookings without deposits contribute disproportionately to cancellations, while non-refundable bookings show near-zero cancellation, highlighting pricing/booking policy impact.

04

LONGER LEAD TIMES INCREASE CANCELLATION PROBABILITY

Bookings made far in advance show consistently higher cancellation rates, indicating speculative or uncertain customer intent.

05

HIGHER PRICING (ADR) IS LINKED TO INCREASED CANCELLATIONS BEYOND A THRESHOLD

Higher pricing (ADR) is linked to increased cancellations beyond a threshold

06

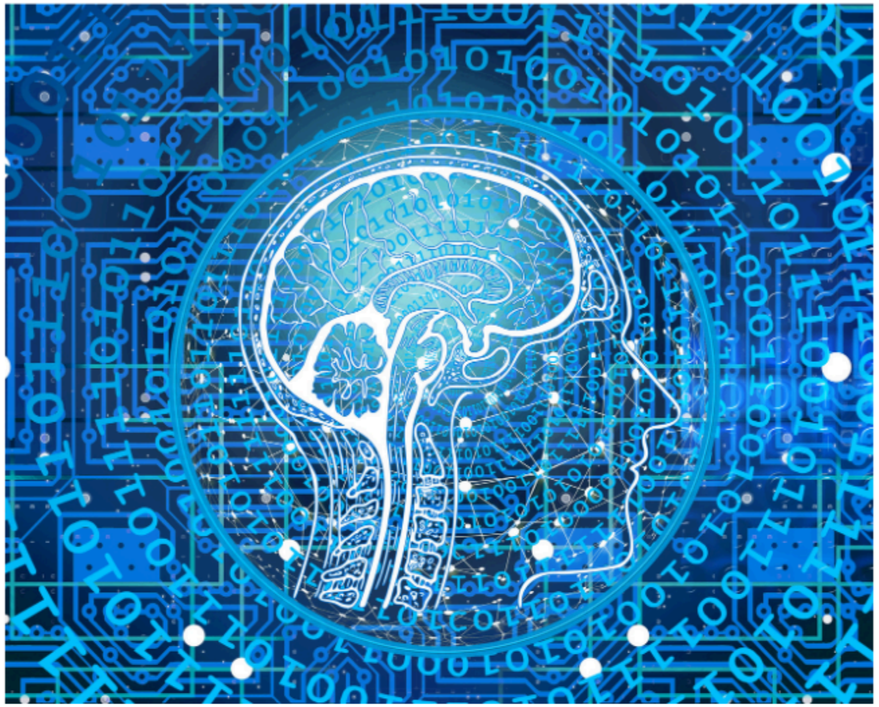
REPEAT CUSTOMERS ARE HIGHLY RELIABLE BUT UNDERUTILIZED (~3%)

A very small portion of bookings come from repeat guests, yet they show significantly lower cancellation behavior, representing a missed retention opportunity.

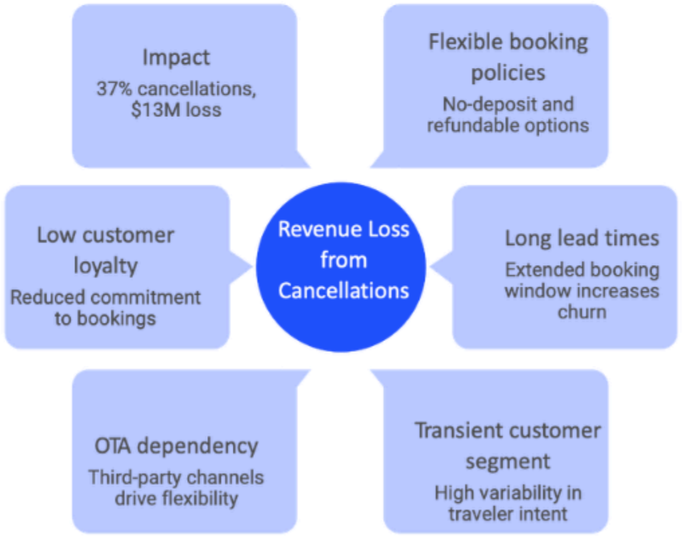
07

ONLINE TRAVEL AGENTS DOMINATE BOOKINGS BUT MAY CONTRIBUTE TO INSTABILITY

Most bookings come from online channels, which are typically associated with higher cancellation flexibility and lower customer commitment.



Advanced Analysis



Statistical & Analytical Work Performed

- Conducted correlation analysis on numerical features to identify relationships between variables
- Used a correlation heatmap to visually analyze patterns and dependencies across features
- Performed data cleaning and preprocessing to ensure reliable statistical results (handling missing values, dropping irrelevant columns)

Key Insights Beyond Basic EDA

- Identified inter-feature relationships that are not visible in simple visualizations
- Improved understanding of which variables are strongly or weakly related
- Enabled better interpretation of data structure and dependencies for further analysis

Tools & Libraries Used

- pandas, numpy → data handling and preprocessing
- seaborn, matplotlib → correlation heatmap and visualization

Recommendations

INTRODUCE TIERED DEPOSIT POLICIES

- Apply mandatory or partial deposits for high-risk bookings (long lead time, city hotels)
- Reduces speculative bookings and last-minute cancellations

DYNAMIC PRICING BASED ON CANCELLATION RISK

- Adjust ADR for high-risk segments (offer incentives for non-refundable plans)
- Balance occupancy vs revenue stability

OPTIMIZE LEAD TIME STRATEGY

- Limit or incentivize shorter booking windows
- Offer discounts for closer-to-date bookings to reduce uncertainty

STRENGTHEN LOYALTY PROGRAMS

- Target repeat customers with exclusive offers
- Increase share of low-cancellation, high-value guests

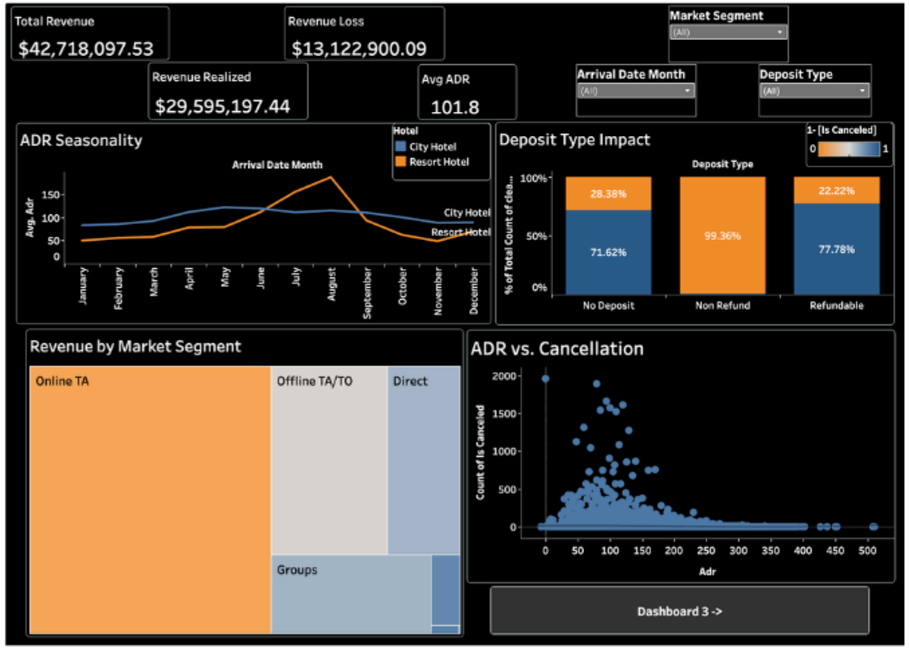
CHANNEL OPTIMIZATION (REDUCE OTA DEPENDENCY)

- Encourage direct bookings via website incentives
- Reduce reliance on channels with high cancellation flexibility

Tableau Dashboard Walkthrough

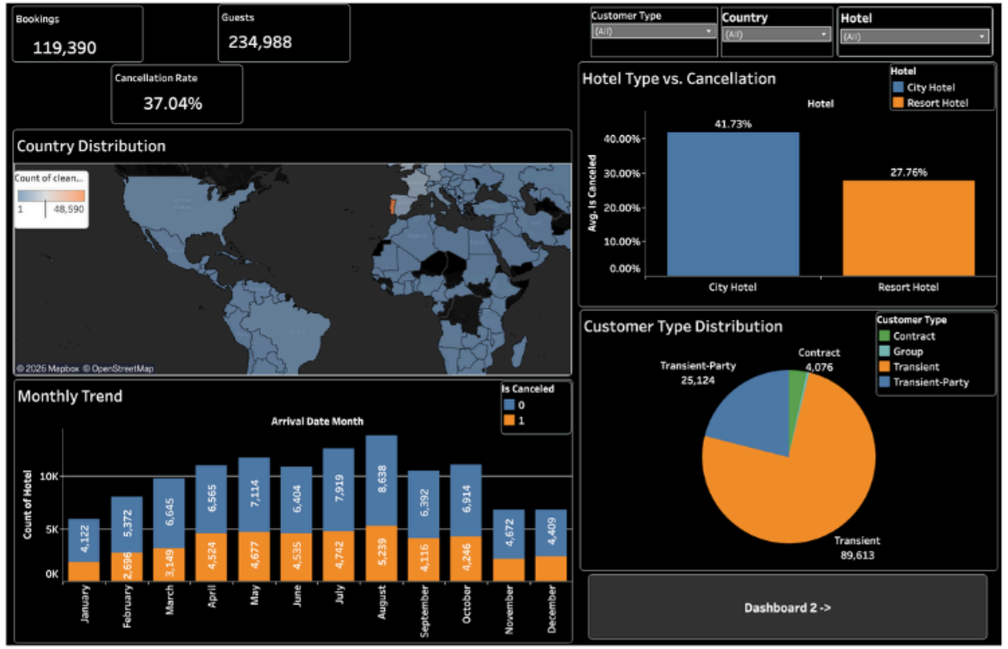
Executive View (For Decision Makers)

- High-level snapshot of Total Revenue, Revenue Loss, Cancellation Rate, ADR
- Quickly highlights business performance and financial risk
- Designed for fast decision-making (CEO-level view)



Operational View (For Analysts)

- Drill-down by Hotel Type, Deposit Type, Lead Time, Customer Segments
- Enables identification of high-risk bookings and cancellation drivers
- Supports root cause analysis and action planning



click - tableau link

Impact Value

Estimated Business Impact

Reduce Cancellation Rate by 8–12%
→ Through deposit policies + pricing optimization

Recover ~$3M–$5M in Revenue
→ From current ~$13M revenue loss

Improve Revenue Realization by ~10–15%
→ More bookings convert into actual stays

Increase High-Value (Repeat) Customers by 2–3×
→ Lower churn, higher lifetime value

Better Demand Predictability
→ Improved inventory & pricing decisions

Cost of Inaction

- Continued high revenue leakage (~35–40%)
- Inefficient pricing → lost profit opportunities
- Overdependence on high-risk booking channels
- No control over cancellation-driven volatility

Cost of Why Act Now?

- Small policy changes (deposits, pricing, targeting) can deliver high ROI without major operational cost

Limitations & Next Steps

Next Steps

1. Build Cancellation Prediction Model
- Use ML to predict booking-level cancellation probability

2. Real-Time Risk Dashboard
- Track live bookings + risk scores for proactive action

3. Integrate External Data
- Add seasonality, holidays, local events, competitor pricing

4. A/B Test Business Strategies
- **Test deposit policies, pricing changes, loyalty offers**

Limitations

- No customer-level behavioral data (income, preferences, intent)
- Limited visibility into external factors (seasonality, events, competition pricing)
- Assumes historical patterns remain consistent
- No real-time tracking → insights are post-analysis

Why these slides are strong:

- Quantifies impact (judges care about numbers)
- Shows clear business value
- Acknowledges limitations (adds credibility)
- Ends with forward-looking execution plan

Thank You!